

Venkat Varadarajan

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WORK EXPERIENCE

Oracle, Seattle, WA — Consulting Member of Technical Staff

Apr 2022 - Current

- Designed and prototyped an AI agent using LangChain, evolving it into a platform capability for semantic ranking of access-controlled data sources
- Led full-stack development of an interactive data lineage platform based on OpenLineage and Marquez
- Built an automated, highly parallelized data processing pipeline that transforms unstructured data sources into a vector knowledge base
- Implemented database-native RAG for natural-language question answering over unstructured data using vector knowledge bases and integrated LLM inference
- Reduced load times of data from cloud storage to lakehouse stores by 40% through optimized object prefixes
- Built a cross-cloud cost attribution, billing, and metering service for a managed database service deployment on AWS

Oracle Labs, Austin, TX — Senior and Principal Member of Technical Staff

Feb 2016 - Apr 2022

- Led development of automated feature extraction and machine learning algorithm selection across classification, regression, and anomaly detection tasks
- Built an auto-tuning engine for machine learning models using a patented gradient-based hyperparameter optimization and asynchronous parallel search approach
- Architected and implemented the control plane enabling zero-downtime, on-demand resizing of the MySQL in-memory query accelerator for live workloads

SELECTED PATENTS AND PAPERS

- *Gradient-based auto-tuning for machine learning and deep learning models*, US Patent 11,176,487
- *Scalable and efficient distributed auto-tuning of machine learning and deep learning models*, US Patent 11,120,368
- *Oracle AutoML: A Fast and Predictive AutoML Pipeline*. VLDB 2020
- *Rapid: In-memory analytical query processing engine with extreme performance per watt*. SIGMOD 2018

LANGUAGES C/C++, Python, Java, SQL, PL/SQL

SKILLS Langchain, AI Agents, RAG, Vector store, ETL/ELT, Iceberg, REST services, AWS, OCI, AutoML

EDUCATION

University of Wisconsin-Madison, — Ph.D., M.S. Computer Science

College of Engineering Guidy (CEG), Anna University — B.E Computer Science